Riyasdeen Samsudeen

HCI & Design Master's Student
Product Development Engineer · Mechatronics Engineer

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Enschede, Netherlands · Espoo, Finland

PROFILE

Master's student in **Human Computer Interaction & Design** (HCID) with **strong technical expertise** in embedded systems, electronics and robotics. Skilled in user research and usability evaluation, as well as in developing hardware from concept to functional prototype to industrial design and testing.

CORE STRENGTHS

- Human-Centred Design
- Hardware + UX Prototyping
- Embedded Systems
- Robotics & Drones
- Product Innovation

SKILLS

Design & Research

User-centred design · Usability testing · User research · Interaction design · Concept design · Figma

Hardware & Engineering

Product development · Industrial Design · 3D modelling · Open-source hardware · PCB Design · Integration · Firmware · Technical illustration · Rapid prototyping · Image processing

Platforms

ESP · ATmega · Raspberry Pi · Jetson · NXP i.MX · Pixhawk

Tools & Code

Fusion 360 · Eagle · Arduino IDE · Embedded C/C++ · Python · HTML/CSS

EDUCATION

University of Twente, Netherlands · 2026-2027
M.Sc ICT Innovation (EIT Digital Double Degree), HCID with Intelligent Systems

Aalto University, Finland · 2025-2027
M.Sc ICT Innovation, (EIT Digital Double Degree) Human-Computer Interaction & Design (HCID)

SASTRA University, India · 2019-2023
B.Tech Mechatronics

University of Aegean, Greece · 2026
Summer School, Maritime Informatics & Robotics

RESEARCH & PROFESSIONAL EXPERIENCE

Finndent Oy - Aalto Product Development Project, Finland — Sep 2025 - May 2026
Project Manager & Engineer

- **Led the redesign of a dental instrument bridge** with two interactive displays, running **user research and usability testing in dental clinics** to improve handling and user experience, alongside electronics, industrial hardware and mechanical integration.
- Designed a custom PCB shield for NXP i.MX. Took the new product design through prototyping to manufacturing, on a **€10,000 development budget**.

Spezar Tech Pvt. Ltd., Thanjavur, India — Feb 2021 - Aug 2026
Founder

- **Built products end to end** in robotics, embedded and IoT from architecture, PCB, firmware, sensors, prototyping, testing to real time deployment.
- Built and led a multidisciplinary team in India. Work recognised by **government and international innovation awards**.

INRO Labs, Chennai, India — Jun 2018 - Nov 2020
Junior R&D Intern & STEM Trainer

- Developed a surveillance robot for COVID-19 zones; trained 500+ school students in electronics, mechanics and embedded tech.

RELATED PROJECTS

AuraSense - Screen-Free Haptic Stress Relief · 2026
Screen-free stress relief device combining touch, haptics and visual interactions. Built the electronics, firmware and interaction logic through iterative user testing.

Semi-Autonomous Visitor Guidance Robot · 2023-2024
Mobile robot platform focused on human-robot interaction and accessibility.

Smart Footwear for Parkinson's Gait Analysis · 2023-2024
Human-centred product design for gait analysis, supporting Parkinson's diagnosis.

Drone-Ready IoT Organ Transport Box · 2023
Hardware for a drone-delivered box that monitors organs and supplies in real time.

IoT System for Remote Farm Operations with ABB VFD · 2023
Remote IoT control of VFD-driven farm equipment with user friendly control dashboard.

Portable Wireless Control System Module for Robotics and IoT· 2023
Modular wireless controller with multi-port sensor inputs for robotics and IoT.

AWARDS & LEADERSHIP

- **One of 10 Fellows from Finland**, BabyVC Nordics & Bloom Labs
- **Mentor**, Imperial College London Global Challenge Lab · **Evaluator**, Smart India Hackathon · **Volunteer**, Slush Finland
- **Tamil Nadu Student Innovator Award**, Govt. of Tamil Nadu, 2020-21 & 2021-22
- **Most Creative Award**, International Space Challenge, Singapore
- **India Top 5**, ActInSpace (CNES, ESA, Airbus) · **2nd Runner-up**, World Robotics Championship